# Robot Procedure 2016_018

April 18th

# 2016

**SERVO MOTOR BRAKE CHECK FUNCTION**
**ROBOT OPTIONAL FEATURE**
**ROBOT PREVENTIVE MAINTENACE TOOL**

[Applicable for R30iA & Above controller ]

## Overview

We are all well aware of checking the brake coils and its specification using external measuring tools at the time of troubleshooting. The Brake check function is built in optional feature in Robot controller model R30iA & Above.

***Brake check function (A05B-2500-J951)*** is a function to check that motor does not move when motor torque is applied under the brake is engaged. This function makes user to cause early detection of brake failure.

This function also informs the check results to user by warning message or system variables.

## Initial setting before use Enable/Disable switch for brake check function

The following system variable is the switch that enable/disable brake check function.

**$BCCFG.$BCK_ENABLE TRUE: Enable (default) / FALSE: Disable**

Please confirm this variable is TRUE (Enable) before use. The default value of this variable is TRUE (Enable). If this variable is FALSE (Disable), brake check never starts.

## Enable/Disable switch for each axis

On this function, user can customize enable/disable brake check for each axis. Please set the following system variable to TRUE (Enable) for the axis where you want to execute brake check.

**$BCK_GRP[group].$CHK_AXIS[axis] TRUE: Enable / FALSE: Disable**

On robot axis, the default value is TRUE (Enable). On except robot axis (aux axis, servo gun, or etc.), the default value is FALSE (Disable).

In addition, if you execute brake check on dual drive axis, please set this variable to TRUE (Enable) for both master and slave axis.

## Setting brake torque

This function needs brake torque for each axis in order to determine torque command during brake check.

Please set the following system variable with the brake torque [kgf*cm] for each axis.

**$BCK_GRP[group].$BRK_TORQUE[axis] brake torque [kgf*cm]**

On robot axis, this variable is set automatically, however, **on except robot axis (aux axis, servo gun, or etc.), this variable is not set automatically, so please manually set this variable with brake torque if you execute brake check on except robot axis.**

**For specification value of brake torque, please refer *AC SERVO MOTOR DESCRIPTIONS "BRAKE SPECIFICATIONS"*. Please note that the unit of this variable is [kgf*cm].**

| AC Servo Motor Description | Manual Specification |
|---|---|
| AC Servo Motor α Series Description | B-65142EN |
| AC Servo Motor β Series Description | B-65232EN |
| AC Servo Motor αi Series Description | B-65262EN |
| AC Servo Motor βi Series Description | B-65302EN |

In addition, if you execute brake check on dual drive axis (except robot axis), please set this variable as follows. (For robot axis, it is no need to set even if it's dual drive axis.)

Case-1: Either master or slave motor has one brake.

**Please set this variable with the brake torque [kgf*cm] for only the axis that has brake. For the axis that has no brake, please set 0.0 (default).**

Case-2: Both master and slave motors have brake respectively.

Please set this variable with the brake torque [kgf*cm] for both master and slave axes. And then the brake torque value of master and slave axis must be same. If these values are different in the Case-2, it is not supported.

## Start brake check

To start brake check, please set the following system variable to TRUE.

**$BCCFG.$BCK_START TRUE: Brake check is running / FALSE: Brake check is finished**

User can start brake check by parameter instruction of TP program or manually setting on system variable screen. This variable automatically returns to FALSE when brake check is finished. Please not to move robot till this variable automatically returns to FALSE. In addition, if brake check function is disabled **($BCCFG.$BCK_ENABLE=FALSE),** this variable immediately returns to FALSE even if this variable is set to TRUE.

Follows is a sample of TP program that execute brake check.

```
1: J P[1] 100% FINE
2: L P[2] 2000mm/sec FINE
3: WAIT 1.00 (sec)
4: $BCCFG.$BCK_START=1
5: WAIT $BCCFG.$BCK_START=0
6: L P[3] 2000mm/sec FINE
```

Brake check cannot be started while the robot is moving as specification, so this program waits for robot to stop completely by WAIT instruction at L3. And this program starts brake check at L4 and waits to finish brake check at L5. As just described on this sample program, brake check must be executed under the robot stops completely.

> **NOTE**
> In case of brake failure, the motor can move during brake check. So please be sure to execute brake check in sufficiently spaced layout. Total motion command per single check is vary by robot or axis, however, it is up to 200,000[pulse]. This is about 0.4[rev] in motor revolution and about 1.4° in degree of the angle assuming the gear ratio as 1/100.

Additionally, in case where the power is shut OFF during brake check for some reason, the following alarm will be posted at next power ON. This alarm indicates abnormal termination at last check.

**BRCH-003 Last check was failed**

The following system variable will be also set to TRUE at next power ON.

**$BCCFG.$LAST_FAILED TRUE: Last check was failed / FALSE: Last check was normally finished**

In this case, the results of brake check will not be generated, so please execute brake check again at next power ON. If next brake check would be normally finished, this variable will return to FALSE automatically.

## Results of brake check

**$BCCFG.$BCK_START** automatically returns to FALSE when brake check finishes. And then the results are also posted as warning message.

If all brake passes as “Good” brake, the following message will be posted.

**BRCH-001 All brake are normal**

By contrast, if brake failure is detected as “Bad” brake, the following message will be posted.
Ex) Brake failure is detected on group1/J2.

**BRCH-002 Brake abnormal (G:1, A:2)**

The results are also recorded on the following system variable at the same time. The value of this variable is kept till next brake check starts.

**$BCK_GRP[group].$CHK_RESULT[axis] TRUE: Brake failure / FALSE: Normal brake (default)**

## Limitations

- Brake check function cannot be executed under the following conditions, or alarm will be posted.
- Calibration has not been done.
- Robot is moving.
- Servo is OFF.
- Servo parameter is under updating.
- Dual motor is under the adjusting mode.
- Even though the axis has brake and **$BCK_GRP[group].$CHK_AXIS[axis] is TRUE, brake torque ($BCK_GRP[group].$BRK_TORQUE[axis]) is not set.**
- Teach Pendant is enabled.
- Any axis is nearby stroke limit when brake check starts.

## Caution

- It takes about 10[sec] for single brake check execution.
- **When the motor is heating up, the motor torque can decrease, so it may not be able to detect brake failure. To detect it with high accuracy, please execute brake check preferably when the motor is in low-temperature, e.g. first thing in the morning.**

## END OF PROCDURE